Lecture Scribe CSE400

L15_L17.pdf
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Lecture Scribe Writing Guidelines.pdf
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> Generate a lecture scribe intended to serve as reference material for exam preparation. Use *ONLY* the provided context (Lecture 15 slides).
> *Instructions:*
> 1. *Purpose:* The scribe must be exam-ready, strictly following the content of the slides.
> 2. *Content:* Include definitions, assumptions, theorem statements, proofs or derivations, and worked examples *exactly* as presented in the lecture materials.
> 3. *Constraints:* Do not introduce new examples, outside information, or hallucinations. Do not use phrases like 'for example' to add your own content.
> 4. *Structure:* Organize the scribe clearly with appropriate headings corresponding to the lecture flow.
> *Goal:* Produce a clean, accurate revision document based solely on the attached file."

Below is the **lecture scribe generated strictly from the Lecture 15 slides**. The content follows the lecture structure and includes definitions, assumptions, derivations, and worked examples exactly from the slides without introducing new explanations or examples.

The lecture slides used as context are from:

CSE400 – Fundamentals of Probability in Computing

Lecture 15: Joint Random Variables and Joint Distributions

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1. Why Study Joint Random Variables?

In science and in real life, we are often interested in **two (or more) random variables at the same time**.

Examples include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

These situations involve **relationships between variables**, which require joint probability models.

Example 1: Smart Transportation

Two random variables:

- X = Vehicle Speed
- Y = Traffic Density

Question:
Can speed be high when the traffic density is also high?

These variables are **correlated random variables**.

Applications in intelligent transportation systems include:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Two random variables:

- X = Signal-to-Noise Ratio (SNR)
- Y = Data Throughput

Throughput depends on SNR, but network congestion may affect this relationship.

Applications:

- 5G scheduling

- Adaptive modulation
- Network planning

Additional Examples

- Weather prediction:
  X = Rainfall, Y = Temperature
- Drone delivery systems:
  X = Wind speed, Y = Delivery time
- Rolling two dice:
  $X + Y = 7$ and $X > Y$ requires a joint PMF.

Importance of Joint Distributions

When multiple random variables are involved:

- A **joint distribution** allows computation of probabilities involving both variables.
- It helps understand **relationships between variables**.
- If variables are independent, analysis is simpler.
- If not independent, **covariance and correlation** measure dependence.

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2. Discrete Case – Joint PMF

Setup

Suppose X and Y are discrete random variables.

$$X = \{x_1, x_2, \dots, x_n\}, \quad Y = \{y_1, y_2, \dots, y_m\}$$

The ordered pair (X, Y) takes values in the **product set**:

$$\{(x_1, y_1), (x_1, y_2), \dots, (x_n, y_m)\}$$

Definition: Joint PMF

The **joint probability mass function (PMF)** is defined as

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This gives the probability of the **joint outcome occurring simultaneously**.

3. Joint PMF Table Representation

The joint PMF can be organized in a table.

X \ Y	y_1	y_2	...	y_j	...	y_m
x_1	$p(x_1, y_1)$	$p(x_1, y_2)$	...	$p(x_1, y_j)$	...	$p(x_1, y_m$
x_2	$p(x_2, y_1)$	$p(x_2, y_2)$	...	$p(x_2, y_j)$	...	$p(x_2, y_m$
...	...	...	...	...	...	...
x_n	$p(x_n, y_1)$	$p(x_n, y_2)$	...	$p(x_n, y_j)$	...	$p(x_n, y_m$

Key Properties

1. $$0 \leqslant p(x_i, y_j) \leqslant 1$$

2. $$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

4. Example: Rolling Two Dice

Experiment

Roll two fair dice.

Define:

- X = value on the first die
- T = total (sum) of both dice

Each outcome (i, j) has probability

$$\frac{1}{36}$$

Thus we construct the **joint probability table of** (X, T).

Observations

- Entries are either 0 or $1/36$.
- X and T are **not independent**.

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5. Continuous Case – Joint PDF

The continuous case is analogous to the discrete case.

Correspondence:

- Discrete values → Continuous intervals
- Joint PMF → Joint PDF
- Sums → Double integrals

If

$$X \in [a, b], \quad Y \in [c, d]$$

then the pair (X, Y) takes values in

$$[a, b] \times [c, d]$$

Definition: Joint PDF

A function $f(x, y)$ is called the **joint probability density function** of X and Y if

$$P((X, Y) \text{ lies in a small rectangle of area } dx\,dy) \approx f(x, y)\,dx\,dy$$

Key Properties

1.

$$f(x, y) \geqslant 0$$

2.

$$\int_c^d \int_a^b f(x, y)\,dx\,dy = 1$$

Note:

The joint PDF $f(x, y)$ **may take values greater than 1**, because it represents a **density**, not a probability.

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6. Example

Given:

$$f(x, y) = 4xy, \quad 0 \leqslant x \leqslant 1,\ 0 \leqslant y \leqslant 1$$

The pair (X, Y) takes values in the **unit square** $[0, 1]^2$.

Tasks:

1. Show $f(x, y)$ is a valid joint PDF.
2. Visualize event

$$A = \{X < 0.5,\ Y > 0.5\}$$

3. Compute $P(A)$.

Step 1: Check Validity (Normalization)

$$\int_0^1 \int_0^1 4xy\, dx\, dy$$

$$= \int_0^1 [\int_0^1 4xy\, dx]\, dy$$

$$= \int_0^1 2y\, dy$$

$$= [y^2]_0^1 = 1$$

Hence

- $f(x, y) \geqslant 0$
- Total probability = 1

Therefore it is a **valid joint PDF**.

Step 2: Compute $P(A)$

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy\, dy\, dx$$

$$= \int_0^{0.5} [2xy^2]_{0.5}^{1} dx$$

$$= \int_0^{0.5} 2x(1 - 0.25)\, dx$$

$$= \int_0^{0.5} \frac{3}{2} x\, dx$$

$$= \frac{3}{16}$$

Thus,

$$P(A) = \frac{3}{16}$$

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7. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

The notation

$$X \leqslant x,\ Y \leqslant y$$

represents the event

$$\{X \leqslant x \text{ and } Y \leqslant y\}$$

Definition

The **joint cumulative distribution function** is

$$F(x, y) = P(X \leqslant x, Y \leqslant y)$$

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8. Joint CDF – Continuous Case

If X and Y are continuous random variables with joint density $f(x, y)$ over $[a, b] \times [c, d]$, then

$$F(x,y) = \int_c^y \int_a^x f(u,v)\,du\,dv$$

Recovering the Joint PDF

Since there are two variables, we use partial derivatives:

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

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9. Joint CDF – Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, then

$$F(x,y) = \sum_{x_i \leqslant x} \sum_{y_j \leqslant y} p(x_i, y_j)$$

Interpretation

- Add all probabilities in the table
- For rows with $x_i \leqslant x$
- And columns with $y_j \leqslant y$

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10. Properties of the Joint CDF

The joint CDF $F(x,y)$ must satisfy:

1. Non-decreasing

$F(x,y)$ is **non-decreasing**.

If x or y increases, $F(x,y)$ must stay constant or increase.

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2. Lower-Left Corner Property

$$F(x,y) = 0$$

at the lower-left of the joint range.

If the lower-left corner is $(-\infty, -\infty)$,

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x,y) = 0$$

3. Upper-Right Corner Property

$$F(x, y) = 1$$

at the upper-right of the joint range.

If the upper-right corner is (∞, ∞),

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$

If you want, I can also **convert this exact scribe into Overleaf-ready LaTeX (copy-paste code)** so you can directly submit it for your **CSE400 scribe assignment**.